

TATHAGATA BANERJEE

Software Engineer, NLP & Agentic AI, Quantum Computing

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SUMMARY

NLP and agentic AI engineer with hands on, end to end ML deployments: a transformer based ticket router, a sentiment classification API, and an agentic retrieval system (RAG) over quantum NLP research papers, each with its own CI/CD and cloud setup. Brings 8 years of software engineering experience in backend development and test automation at Dell Technologies and Accenture, including exposure to applied ML and NLP. Completed a research internship at CDAC building an original hybrid quantum-classical architecture for text classification using lambeq and PennyLane, with a survey paper accepted for poster presentation at an IEEE conference at IIT Patna.

TECHNICAL SKILLS

NLP & Transformers: HuggingFace Transformers, Datasets, Accelerate, Evaluate, DistilBERT fine-tuning, weak supervision labeling, sentence-transformers, cross-encoder reranking

LLM & Agentic Systems: LangChain (LCEL), LangGraph (StateGraph, ToolNode, conditional routing), RAG (Retrieval-Augmented Generation) pipelines, multi-query retrieval, Qdrant, model evaluation across Groq and OpenRouter providers

Quantum Computing & QNLP: Qiskit, lambeq, PennyLane, DisCoPy, IQP ansatz, Quantum Algorithms

ML Engineering: PyTorch, TensorFlow, Keras, scikit-learn, pandas, evaluation harness design, dataset curation

MLOps & Infrastructure: Docker (multi-stage builds), FastAPI, Uvicorn, Nginx, AWS EC2, AWS S3, GitHub Actions CI/CD, GitHub Container Registry, IAM instance roles, Git LFS

Frontend & AEM: Adobe Experience Manager (AEM), HTML, CSS, JavaScript, AEM Clientlibs

Automation & Testing: Python, Selenium, Cucumber, Java, PyTest, TestNG, RestAssured, Postman, JMeter, Blackduck, Checkmarx

PROJECTS

QNLP Research Agent, RAG + LangGraph

- Built a retrieval augmented Q&A agent over quantum NLP research papers that backs up its answers with paper and section citations, plus tools that generate lambeq string diagrams and IQP quantum circuits from any input sentence
- Retrieval stack: BAAI/bge-m3 embeddings in Qdrant, MMR retrieval (k=15), multi-query retrieval, cross encoder reranking; PDF ingestion with header aware, LaTeX/table safe chunking. Agent built in LangGraph, routing between the RAG tool and three lambeq based tools
- Built a 26 question evaluation set across 3 papers; caught a Unicode bug worth about 11 points of measured recall, then found the top keyword-recall model was fabricating citations on correctly retrieved context, fixed by switching to a more reliable model
- Traced a 55 minute agent stall to a malformed function call bug on the provider side, and a tool misrouting issue to duplicate tool docstrings

Hybrid Quantum-Classical NLP for Emotion Classification (CDAC Research)

- Designed and implemented HybridQuantumLSTMMModel, an original PyTorch architecture combining parameterized quantum circuits (lambeq, PennyLane) with a classical LSTM head, validating full gradient flow from classification loss back to the quantum circuit parameters via the parameter-shift rule. Achieved 81 to 93% accuracy on binary text classification
- Built the full pipeline converting sentences into quantum circuits (CCG parsing, DisCoCat diagram simplification, IQP ansatz compilation; over 1,500 circuits, close to 3,900 trainable parameters), benchmarked across 4 datasets and 2 quantum simulation backends
- Diagnosed a multi-label evaluation pitfall where a headline accuracy figure masked near zero recall on most emotion classes, and traced a later generalization failure to a compounding data attrition cascade and an autograd graph detachment that had silently cut off gradient flow to the quantum parameters
- Built diagnostic tooling including a gradient flow verification harness, cut experiment iteration time from about 2 hours to seconds via caching, and documented the full theoretical and experimental process in a detailed technical reference

ML Based Support Ticket Router: Live: 13.48.162.241/lead-routing

- Independent, personal rebuild of an ML based ticket routing concept similar to a system used at Dell Technologies, built end to end with open source tools
- Fine-tuned DistilBERT for 4-way ticket classification (Sales, Support, Billing, Customer Service), gated by a pretrained spam classifier. Tickets that don't match any known category are rejected instead of force routed, using a dual layer confidence and entropy check
- Found and fixed a data leakage bug in the augmentation pipeline (train/test split ordering) and a domain mismatch issue where a spam classifier trained on corporate email was misclassifying real support tickets
- 78.2% accuracy and 0.78 weighted F1 on a held out 4-class test set (about 20K tickets, random baseline is 25%)

- Deployed on a single free tier EC2 instance running two containerized transformer services together. CI/CD keeps code deploys separate from model weight updates (Git LFS to S3, bind mounted at runtime)

Tweet Sentiment Classifier: Live: [13.48.162.241/twitter](https://twitter.com/13.48.162.241)

- End to end sentiment classification service (FastAPI, web UI, JSON API), fine-tuning DistilBERT for CPU only inference on a 1GB instance
- Used a weak supervision labeling pipeline (VADER derived labels). 90.8% validation accuracy on held out data
- Focused on production concerns: model weights are kept separate from the deploy image (Git LFS/S3) so weight updates don't need a rebuild. About 13ms response time once loaded, zero OOM kills in production on a shared 1GB instance

NLP Auto-Tagging & FAQ Chatbot for Digital Assets

- Building a POC that suggests tags for digital assets from title and description, combining KeyBERT keyphrase extraction with embedding similarity matching against a tag taxonomy, served through FastAPI with a form page
- Adding a lightweight FAQ chatbot to the same form page

PROFESSIONAL EXPERIENCE

Research Intern, CDAC Kolkata / Sept 2025 to Feb 2026

- Designed and implemented a custom hybrid quantum-classical architecture (parameterized quantum circuits via lambeq and PennyLane, combined with a classical LSTM) for text emotion classification, validating end-to-end gradient flow from classification loss through to the quantum circuit parameters, and benchmarked it across 4 datasets and 2 quantum simulation backends (see Projects for detail)
- Worked on a DWG to Graph conversion pipeline for AutoCAD geometric entities, including feature extraction for Graph Neural Network (GNN) compatible node and edge attributes
- Co-authored a survey paper, "A Survey of Quantum Natural Language Processing, From Foundations to Near Term Applications." Accepted for poster presentation at an IEEE conference at IIT Patna
- Worked with mentors and cross functional teams to benchmark experimental datasets and validate research outcomes

Dell Technologies / Oct 2018 to March 2025

Senior Software Engineer / Apr 2023 to Mar 2025

- Security champion for the team: ran vulnerability assessments with Blackduck and Checkmarx and coordinated mitigation, addressing over 91% of identified vulnerabilities
- Tested and validated an NLP chatbot/FAQ recommender proof of concept built for AEM forms, evaluating response accuracy and conversational flows
- Mentored junior engineers on automation practices and security awareness

Software Engineer 2 / Jul 2021 to Apr 2023

- Worked on AEM front end development (HTML, CSS, JavaScript, Clientlibs): template and component updates, accessibility improvements
- Contributed to backend services for a Microservices/Spring Boot project
- Maintained and extended Python/Selenium automation frameworks used in CI/CD

Software Engineer 1 / Oct 2018 to Jul 2021

- Built automated test suites (Selenium, Cucumber, Java) integrated into GitLab CI/CD, reducing manual testing effort
- Tested an ML based lead routing classification system end to end (2020), which later became the basis for an independent personal rebuild (see Projects)
- Ran API test automation (Postman, RestAssured) and maintained GitLab CI/CD pipelines
- Coordinated cross team defect triage. Worked as SDET for AEM front end in test across multiple applications

Associate Software Engineer, Accenture / May 2017 to Oct 2018

- Executed manual and automated testing within Agile/SCRUM teams to find defects early
- Helped develop test strategies and test plans. Supported CI environments and reported results to stakeholders

CERTIFICATIONS

Joint AI and Quantum Computing (Pursuing), IISc Bangalore & QpiAI | Certificate Course in AI/ML, CDAC Kolkata, Aug 2025 | Quantum Computing & Programming (QBronze), QWorld, May 2025 | Quantum Computing, C-DAC & IIT Roorkee, May 2025 | Basics of Quantum Information, IBM, Apr 2025 | Quantum Business Foundations, IBM, Mar 2025 | Certified Ethical Hacker (CEH), EC-Council, Jan 2015 | CCNA, Indian School of Ethical Hacking, Aug 2015

EDUCATION

- B.Tech, Electrical Engineering, KIIT University, Bhubaneswar (2017) | Chartered Engineer (Apr 2018) & Associate Member, AMIE (Mar 2018), Institution of Engineers (India)

RESEARCH

- Survey Paper (WIP): "A Survey of Quantum Natural Language Processing, From Foundations to Near Term Applications"